

Q.1) Language Overview

Ans: Name:-

My language name is Italo.

Theme:-

A minimalist, different syntax language inspired by C++ for creativity prototyping. It uses Italian language instead of English for a personalized and elevated feel.

Purpose:-

To create a differentiation and limelight for those who want to be different and like to know about other languages as well. It has almost all concepts of programming close to C++ without complexity of full syntax which enables faster deployment.

Q.2)

Ans:

Keyword	Equivalent	Example Usage
SE	if	SE($x > 0$)
ALT	else	ALT{y}
MEN	while	MEN($x < 10$) {x+1}
RIT	return	return a+b;
VER	true	bool flag = VER;
FAL	false	bool flag = FAL;
E	and	SE(a E b)
O	or	SE(a O b)
IN	int	IN x = 10;
PRIN	main	IN PRIN()
OMPI	break	OMPI;
CON	continue	SE(x < 0) { CON; }
NO	null	RIT NO;
FL	float	FL x = 0.2;
CR	char	CR my = 'G';

Q.3)

Ans,

Category	Symbol	Description
Operator	+	Addition
Operator	-	Subtraction
Operator	*	Multiplication
Operator	==	Equality comparison
Punctuation	(	Opening Parenthesis
Punctuation	)	Closing Parenthesis
Punctuation	{	Block Start
Punctuation	}	Block End
Punctuation	;	Statement terminator

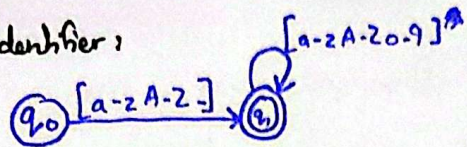
Q.4) Regex

Ans:

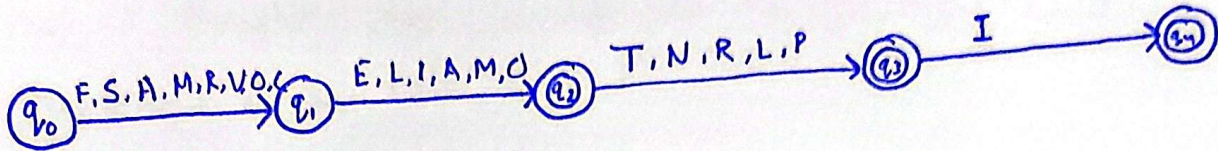
Token Type	Regex	Example
Identifier	$[a-zA-Z_][a-zA-Z0-9_]*$	variant2
Keyword	$[SE\ ALT\ MEN\ RIT\ VER\ FALL\ OMPI\ CON]^+$	SE
Operator	$[+ \ - \ * \ ==]^+$	++
Constant	$[0-9]^+$	49
Separator	$[(){}{};]$	{
Special Character	$[@ \ \$ \ \% \ \#]$	@

Q.5) Finite Automata

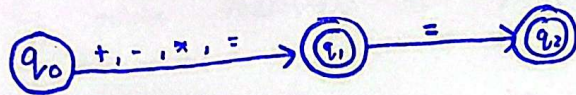
Ans: Identifier:



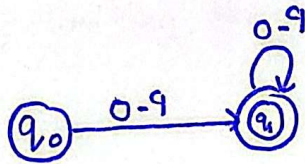
Keyword:



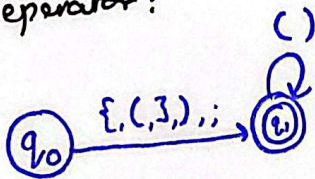
Operator:



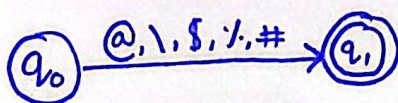
Constant:



Separator:



Special Character:



Q.6)

Ans: I chose my keywords (inspired) from Italian language to make it unique than others and also I have interest in different languages and I also watch Netflix on different translations. So, I thought this would be a unique and interesting thing to do.

My language is unique as we know English is the main language so different language specially this adds freshness to language, and expressiveness. It is similar to C++ so people familiar with it could pick up.

By limiting the words I increased the simplicity.
